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Accurate Wellbore Placement with Rotating Continuous Surveys while Saving Time and Cost in Non-Conventional Shale Play

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Abstract

Drilling Departments aim to drill wells faster, with improved hole quality, and in a safe manner. Wellbore surveying is an important aspect of delivering wells safely. More specifically wellbore surveying should allow:

1. To drill wells without colliding into other wells.
2. To measure a wellbores position accurately enough, that in case of a blowout, a relief well can be drilled quickly and intercept the blowing well at the desired depth, where the accuracy (i.e. positional uncertainty) is transferable between different operators and different service companies.
3. To perform the surveying and anti-collision (AC) calculations in a straightforward and repeatable manner

For the past 30 to 40 years, 6-axis Measurement While Drilling (MWD) tools have been providing downhole surveys that allowed estimating the actual wellbore trajectory. Since the 1990s the same fundamental MWD tool technology has largely been adapted between vendors. The difference has been in procedures and calibration. It was known that the measurements were not without uncertainty, but initially there was no single mutually agreed way to describe and quantify the uncertainties.

The SPE Wellbore Positioning Technical Section (WPTS), also known as the Industry Steering Committee on Wellbore Survey Accuracy (ISCWSA), was founded in 1995 by a group of operators and service companies to agree on a common Positional Uncertainty Model (PUM) that could describe a common way of calculating the Ellipses of Uncertainty (EOU) describing the accuracy of a generic MWD survey. This resulted in the publication of the SPE paper 67616 in 2000 ([Williamson 2000](#)). The ISCWSA briefly describes its purpose as:

"The Industry Steering Committee on Wellbore Survey Accuracy (ISCWSA) was founded to dispel the confusion and secrecy commonly associated with wellbore surveying. And to enable the industry to produce consistent, reliable estimates of survey-tool performance in today's wells. We achieve these goals

through the production and maintenance of standards covering the construction and validation of tool error models" (ISCWSA 2025a).

"In general, ISCWSA does not define or supply the detailed magnitudes (error models, tool codes or IPM files) for a particular survey tool. It is our position that the tool provider is the one who is best placed to assess the uncertainty when using their equipment. Therefore, users should request tool codes from their tool provider. ISCWSA does not supply, check, certify or in any way warrant such error models created by tool providers." (ISCWSA 2025b).

What is commonly referred to as the ISCWSA MWD error model, is in fact a generic conservative model. However, using the huge amount of MWD survey data available to most operators, most large vendors' MWD tools—when the correct procedures are followed—have been shown to qualify for the error model. The raw survey data and the algorithms are also commonly shared, such that third parties can provide Quality Assurance (QA) and Quality Control (QC) for the surveys.

What hitherto has been referred to as MWD surveys, will in this paper be referred to as static-six-axis-MWD-surveys (SSAMWD). These surveys are not acquired during drilling operation, but during a halt in the drilling operation, where the bit is pulled off bottom and the drill-string is stationary. Contrary to this, a continuous-6-axis survey that is acquired during rotation of the drill string and while actively drilling, may be called "Continuous-rotating six-axis-magnetic measurements" (CRSAM). This type of continuous-6-axis survey was proposed (Lowdon 2019) and executed survey cases presented (Elgizawy 2020, Herrera Anda 2024).

What the continuous 6-axis survey lacked was a commonly agreed PUM. Following a pragmatic comparison of SSAMWD and CRSAM surveys (Dreisig and Reynaud 2023), the work on an ISCWSA PUM was kicked off at the 59th ISCWSA meeting (Willerth 2024). A preliminary PUM for 6-axis continuous surveys was presented at the 60th ISCWSA meeting (Willerth and Akklestad 2024) that as time of writing this paper has turned into a release candidate (ISCWSA 2025c). For a vendor to justify using the error model, they need to provide sufficiently large datasets to show that their tool and procedures qualify for using the proposed PUM. Without appropriate confidentiality agreements, it can be hard to share these datasets. What is required is a method to show that the continuous six-axis surveys for a specific tool and wellbore can be shown via statistical analysis of the survey data, to be likely to qualify for the proposed PUM.

This paper presents results from field tests executed on onshore horizontal wells in a non-conventional play in an Argentinian field, with a history of over 50 analogous wells already drilled. Furthermore, the paper outlines an approach for qualifying CRSAM survey performance in real-time during drilling operation. The paper demonstrates that CRSAM surveys offer a method to achieve this objective with quantifiable uncertainty and demonstrate the applicability of CRSAM surveys as a definitive reference to drilling automation algorithms.

Introduction

Directional measurements through magnetic-based techniques can be categorized into three classes:

1. **Static Six Axis Measurement-While-Drilling** (SSAMWD) surveys passing quality control (QA/QC) are associated with an ISCWSA / SPE WPTS Positional Uncertainty Model (PUM) enabling quantifiable Collision Avoidance (CA) calculations. Acquisition typically requires 5 to 15 minutes ensuring pipe standstill and data transmission.
2. **Continuous Rotating Six Axis Measurement** (CRSAM) are acquired while drilling with the bit rotating on bottom and subjected to QA/QC procedures with an associated ISCWSA PUM currently being finalized.
3. **Rotary Steerable Steering unit directional measurements** (RSSUDM) are acquired continuously during drilling with the bit on bottom with a primary objective to continuously enable manual and

automated steering adjustments. The data does not necessarily conform to the quality criteria that enable the application of an ISCWSA PUM.

Acquisition Of Different Survey Types

Since SSAMWD require the drill string (DS) to be static, they are typically performed in conjunction with making connections between adjacent pieces of drill pipe at the rig. Depending on the tool, they may require mudflow down the drill string to drive a turbine which powers the MWD tool. The specific tested tools with CRSAM capability included a battery. This meant that they were able to perform surveys during connections when the pipe sat in the slips and a new joint was made up. This implies the acquisition of SSAMWD surveys at close to no excess rig time. For the surveys reception and qualification at surface more time is required than the bare acquisition window. The pumps are brought up after connections while running back to bottom. Subsequently to establishing flow, a certain transmission time for the mud pulse telemetry is required to send the survey data. If procedures require to halt drilling pending a successful survey decoding and qualification, these steps can contribute additional delays due to directional surveying.

In contrast CRSAM are taken continuously while drilling, and every 3 minutes a survey is telemetered to surface. Once the survey is decoded at surface it gets tagged with time and depth. This also means that for those CRSAM surveys that are only recorded in memory mode, the depth tagging still must be done when the tool memory is retrieved on surface. The CRSAM survey data set is derived downhole from rotational averaging utilizing all six MWD measurement axes and results in five variables that encode the tools drilling direction. At a high level this reduction in degrees of freedom can be understood by a comparison with a SSAMWD survey. For a SSAMWD survey, the six measurement values represent an encoding of the tool orientation, i.e. tool-face, inclination, azimuth, and the reference measurements, total magnetic field, total gravity field, and magnetic dip angle. For a CRSAM survey the rotational averaging effectively removes the tool-face degree of freedom. This gives rise to the implicit advantage that any error sources whose influence on the survey can be randomized by covering a sufficient tool-face spread in SSAMWD surveying, can be considered strongly suppressed in CRSAM surveying. Operationally, CRSAM surveying offers a plethora of solutions and benefits (Table 1).

Table 1—Operational challenges and solutions offered by CRSAM technology capabilities. While some challenges like ROP maximization and tortuosity control are universal to drilling operations, others, such as the mitigation of magnetic mud influence, are very situational. All challenges are already considered during the planning stage and addressed procedurally. CRSAM surveying technology adds a new method to achieve operational targets and provides another layer of possible mitigation options.

Operational Challenges	CRSAM Capability	Operational Benefit
ROP maximization or rig time minimization	Definitive survey while drilling	Removes static survey induced NPT/ILT while maintaining HSE and collision avoidance compliance
Improved tortuosity control	Definitive high-resolution surveying w/o additional delay	Improved steering information leads to improved casing runability
Wellbore placement control	Definitive high-resolution surveying w/o additional delay	Improved manual and automated closed-loop steering decisions
(Increased) Risk of stuck pipe	Definitive survey while drilling	Removes potential flow-off requirement for survey acquisition
Mitigation of battery failures and battery temperature range limitations	Definitive survey while drilling	Delivers flow-on survey
Temporary telemetry decoding errors and pump noise on start up	Definitive survey while drilling	Avoids the need for survey telemetry during pump start-up windows
Interference from magnetic mud contamination and formation wash-out	Definitive survey while drilling	Reduces sediment and cuttings settling during survey acquisition

The RSSUDM are an essential contributor to automated directional drilling (ADD) calculations, offering the forwardmost position of surveying the directional information shortly behind the drill bit, but are subsequently exposed to larger and less controllable error

sources than the dedicated MWD tool traditionally positioned further up. As such, the data quality cannot be expected to meet the strict QC criteria that are employed to derive definitive directional surveys from dedicated MWD tools without additional cost, effort and technical restrictions placed upon adjacent components of the drill string.

All the above survey types must be integrated with the LWD program and must fit within the data bandwidth of the required telemetry-frames. In the case of the test wells the LWD requirement was Azimuthal Gamma Ray (GR) measurements at a density of one per foot (three per meter). With the achieved rate of penetration and the allowable data bandwidth of the telemetry frames, this resulted in a CRSAM average survey spacing of 2m, aligning with the requirement of a directional survey measurement every 2m to 3m.

For the ISCWSA MWD Rev 5 PUM ([ISCWSA 2019](#)), surveys are penalized if the survey interval is greater than 30m / 100ft. It is also a requirement not to mix survey types, and a recommendation not to switch back and forth between PUM during a run to prohibit an overly optimistic uncertainty calculation by reinitializing systematic error sources. This is only warranted when the physical MWD tool is exchanged, for example between runs, but not if it is done several times with the same tool assembly.

The Test Program

This specific field was chosen to offer test wells in a non-conventional play that follow near identical trajectories with only small variations in depth, implying good comparability between different runs. Since the play is a tight reservoir, it poses no Health-Safety-Environment (HSE) or collision risk in the horizontal reservoir sections. The reservoir is tight, requiring large-scale fracturing to produce, and neighboring wells are geologically separated, with clear LWD-detectable boundary layers. The test of automated directional drilling was targeting to standardize and optimize for consistency and speed. The wells architectures and drilling programs are described in [Table 2](#), giving an overview of the drilled runs and utilized equipment. The scope of the compiled test program can be broken down as follows:

- During the drilling of well 1, directional drilling automation is intended to optimize directional drilling decision-making by using RSSUDM to follow the planned trajectory, while relying solely on SSAMWD to provide surveys with a quantifiable PUM.
- In well 2 CRSAM surveys would be acquired for post-run data analysis and PUM validation ([Ali 2020](#)), using Relative Instrument Performance (RIP) test and χ^2 -tests. The analysis should rely on surface available survey data with data retrieved from tools memory as a backup.
- Well 3 was planned so, that SSAMWD surveys would be acquired and posted as a definitive survey at every connection until three stands into the horizontal section, while CRSAM surveys would be gathered at surface while drilling. The combined survey data from SSAMWD and CRSAM surveys would be sent to a specialist and within the hour an on-demand validation using RIP and χ^2 -tests would be provided. Upon establishing agreement of the CRSAM data with the SSAMWD and with the assigned PUMs the horizontal section would be drilled using the CRSAM surveys while only waiting for decoding of SSAMWD after connections, to ensure a successful SSAMWD survey surface reading at least every 10 stands.

Table 2—Test wells by sequence, their architecture and planned wellbore surveying program.
Survey frequency of once per stand is abbreviated as OPS. Measured depth is abbreviated as MD.

Well No	Hole Size	From MD	To MD	Inc.	BHA description	ISCWSA PUM	Survey Frequency
1	12.25 in	10 m	632 m	0°	9.625 in Casing While Drilling	Gyro Rev5	OPS
	8.75 in	632 m	1743 m	0°	8.75 in Mudmotor BHA	MWD Rev5	OPS
	6.75 in	1743 m	2997 m	0-87°	6.75 in Mudmotor BHA	MWD Rev5	OPS
		2997 m	6037 m	+/- 90°	6.75 in Rotary Steerable System, without capability of taking CRSAM surveys	MWD Rev5	OPS
2	12.25 in	10 m	630 m	0°	9.625 in Casing While Drilling	Gyro Rev5	OPS
	8.75 in	630 m	1743 m	0°	8.75 in Mudmotor BHA	MWD Rev5	OPS
	6.75 in	1743 m	5970 m	+/- 90°	6.75 in Rotary Steerable System, with CRSAM capability	MWD Rev5	SSAMD OPS & CRSAM every 2min for comparison
3	12.25 in	10 m	624 m	0°	9.625 in Casing While Drilling	Gyro Rev5	OPS
	8.75 in	624 m	1749 m	0°	8.75 in Mudmotor BHA	MWD Rev5	OPS
	6.75 in	1749 m	+/-3000 m	0-90°	6.75 in Rotary Steerable System, with CRSAM capability	MWD Rev5	SSAMWD OPS & CRSAM every 2 minutes
		+/-3000m	6019m	+/- 90°		CRSAM	CRSAM every 2 minutes & SSAMWD by memory OPS

Automated Directional Drilling

Automated Directional Drilling (ADD) was introduced to the directional drillers while drilling the test wellbores. The key principles of ADD are described in (Hansen 2020): The ADD system is designed to unburden the directional driller by automating routine tasks such as checking the distance to well plan, predicting the wellbore trajectory ahead, as well as deriving steering parameters suitable for an RSS BHA. The system monitors the wellbore trajectory in real-time and adjusts the trajectory through sending steering downlinks triggered by the directional driller in advisory mode. The underlying algorithm is solving a constrained optimization problem for prediction of the wellbore trajectory in real-time. ADD utilizes posted directional surveys that can either be obtained as SSAMWD or CSRAM, in conjunction with near-bit inclination and azimuth measurements from the steering unit of an RSS BHA (i.e. RSSUDM). ADD comes with a user interface for the directional driller that allows to supervise the drilling of the wellbore trajectory or control the wellbore trajectory directly.

Well 1: Assist in optimizing the directional drilling decisions

While drilling the first wellbore the directional drillers have been exposed to the automation technology and could familiarize themselves with the automation system. ADD was fully installed and was receiving data in real-time. ADD requires an RSS BHA for steering the wellbore trajectory. However, the first two runs used a bend motor BHA. Thus, only the data feeds to the automation system could be checked, but no steering proposals could be generated. The third run was done using an RSS BHA and ADD was able to provide steering proposals.

Well 2: CRSAM introduction and post-run validation while also applying ADD

More than 4200 m of the second well have been drilled with an RSS BHA. The ADD system was fully connected and produced steering proposals throughout the run. The steering proposals have been judged by experienced directional drillers for their appropriateness and presented to the directional driller responsible for drilling the wellbore: The steering proposals deemed to be appropriate in almost every situation. Only in rare situations with intermittent network connectivity data loss degraded the quality slightly. Nonetheless, the local directional drillers used only a few of the system's steering proposals as they just started to

familiarize with the system. In other words, ADD was used mainly as an advisory system. From a data perspective, only SSAMWD surveys and near bit information (RSSUDM) were utilized. CRSAM data was not used but recorded for post-run validation.

Well 3: CRSAM on-demand validation and CRSAM informed automated directional drilling

While drilling the 4th wellbore CRSAM survey data has been recorded and carefully compared against static surveys. In fact, upon receiving a positive go-ahead the CRSAM surveys have been posted by the MWD engineer for the remaining 2700 m of drilling and as such were consumed by the ADD system to guide the algorithm. Throughout the total 4500 m of drilling both SSAMWD surveys and CRSAM surveys have been used for prediction of the wellbore trajectory. The appropriateness of the steering proposals judged by an experienced directional driller have been similar as for wellbore 2. However, the local drilling crew used more steering proposals as they gained trust in the automated directional drilling system. Approximately 60% of the wellbore have been drilled with ADD following the automatically generated steering proposals. The increased acceptance emphasizes that carefully "managing the acceptance with management-of-change process with end users" (Torres Cabrera 2024) is crucial for successful introduction of automation and automated workflows.

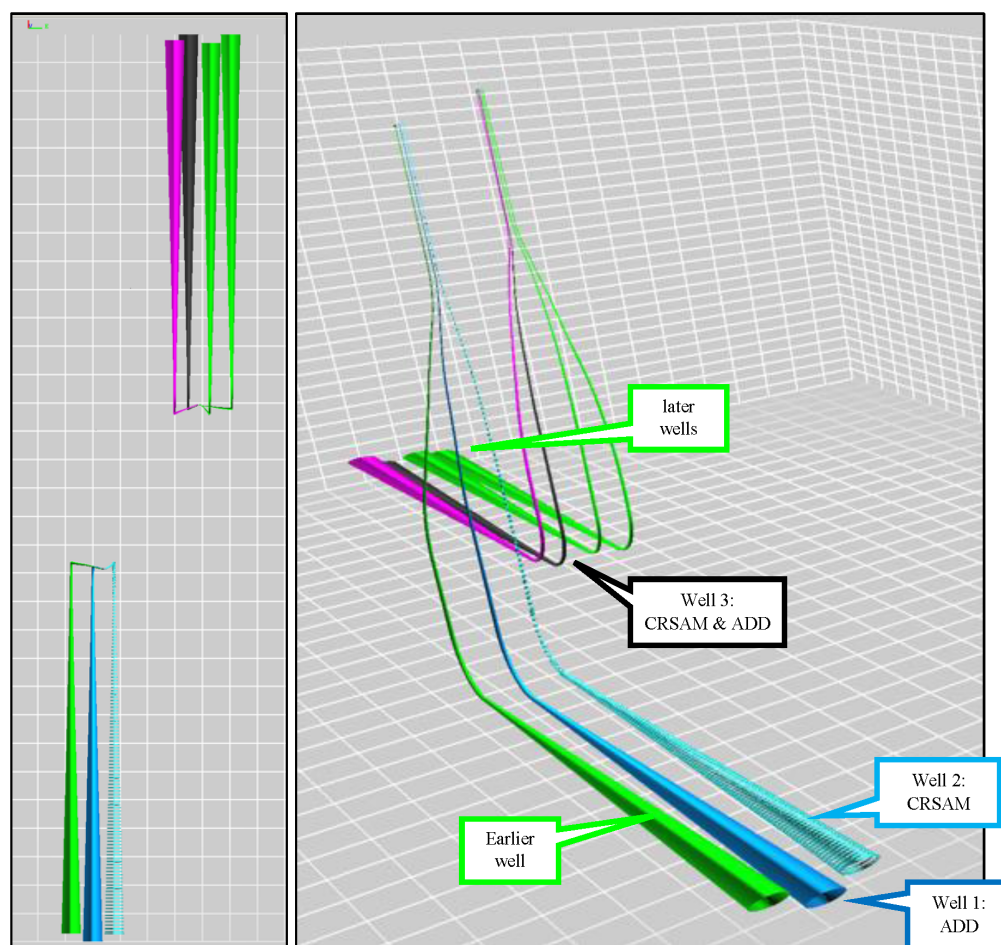


Figure 1—Map view (left) and 3D view (right) of the test wells with wells before and after (in green). Not to the same scale, shown with 3σ EOU.

Continuous Rotating Six Axes Measurement Surveying

The presented test program includes a tiered approach to phase in the operational use of CRSAM surveying data. Firstly, a post-run validation of the surveying modelling assumptions on the last run of well 2 was

carried out to provide exemplary indications of whether the CRSAM and SSAMWD agree within their joint uncertainty margins. Secondly, an on-demand validation during drilling of the last run of well 3 was provided to assure that no violation of the modelling was detectable specifically for this run, to determine a go-ahead-criterion to switch survey operation modes for providing definitive posted surveys using CRSAM for the remainder of the run.

Survey Comparison Methodologies

The SSAMWD surveys are compared against their CRSAM counterpart to provide insights into the relative performance and assess the conformity of the survey data to the underlying uncertainty modelling assumptions that are used to derive positional uncertainties. Survey comparisons can be performed with varying levels of detail, ranging from relative instrument performance (RIP) tests to coordinate difference and χ^2 -tests (chi-squared tests). To compare surveys at a given depth, the higher density survey might be interpolated to match the position of the lower density survey. This interpolation renders coordinate-difference testing on the along hole component uninteresting, as any depth error is transferred to the other coordinates by this procedure. It is important to note, that all these tests constitute a necessary condition for conforming with the error modelling assumptions, none of them can provide a sufficient condition.

Relative Instrument Performance Tests

The RIP test is a broadly applied heuristic gauge of joint uncertainty modelling conformity of the analyzed data. It utilizes the average and standard deviation of the differences in measured inclination and azimuth between two surveys. These mean and standard deviation values are divided by the 1-sigma result of the joint error model contributions to the inclination and azimuth differences. Depending on the resulting numeric value of this ratio, a categorical performance judgement based on empirical data is passed ([Ali 2020](#), [Naschenveng 2023](#)) for the systematic and random error performance separately, by considering the normalized average and standard deviation, respectively. The categorization is based on empirical data, that in turn implies a certain combination of uncertainty models under comparison. Thus, this categorization might not be suitable to fully validate the differential performance of SSAMWD and CRSAM surveys. In turn, a more quantitative test, that can be linked to significance intervals is desirable.

Relative Instrument Performance Quality Control (RIPQC)

While the RIP test on its own does provide a qualitative (i.e. semantic) measure of relative performance, based on quantitative analysis, it still relies on a heuristic that gauges the observed deviations against the full error modelling performance. An alternative approach is to utilize a separation of contributing error terms into systematic and random, and only compare the deviation average and standard deviations against the respective error modelling contributions, while still considering the estimation uncertainty of the average. In this manner, a desired level of confidence can be prescribed to cover the cumulative likelihood of the observed results being compatible with the modelling assumptions. In more plain terms, it attempts to answer the question: "Is the observed systematic deviation, including its estimation uncertainty range of 99.7% fully enclosed within the 99.7% expectation interval for systematic errors derived from the uncertainty modelling assumption?". Proceeding similarly for the observed standard deviation leads to acceptance limits that are determined via the 99.7% cumulative probability of a χ^2 distribution of the sample variance that depends on the number of included observations (i.e. samples or survey stations) and the expected random variance on the survey differences informed by the respective uncertainty modelling assumptions. Again, in more plain terms, it attempts to answer the question: "Is the observed standard deviation compatible with the error modelling assumptions, given the number of survey stations under consideration, within a margin that covers 99.7% of expected cases?". This approach can provide a simple pass or fail criterion while simultaneously allowing to set confidence level expectations as per desired application.

χ^2 -Tests

This method tests the cumulative squared normalized deviations between two surveys against the 99.7% confidence interval of the underlying χ^2 distribution. It assumes a mean-free normal distribution as the origin of the observed deviations. On the one hand, this implies that it can only be applied in the absence of systematic contributions to the observed deviations, e.g. when comparing MWD surveys to gyro surveys, the MWD must be free of magnetic axial interference (Ali 2020). On the other hand, it does attempt to cover the systematic errors by including them into the normalization procedure. A pragmatic heuristic is to perform the test only join sections of the survey log including 15 consecutive surveys stations (Ekseth 2007). This heuristic can successfully mitigate the cumulative influence of systematic errors in the test but is not guaranteed to succeed. When it is required to perform a survey comparison on a specific run, even in the presence of systematic errors, this sensitivity to systematic errors can lead to two distinct and equally undesirable scenarios:

1. It may ignore the presence of systematic errors and have the test fail due to them overly biasing the calculation. This constitutes a conservative failure, since the systematic error sources are accounted for in the position uncertainty modelling and the test is simply unsuited to determine whether the survey data is conforming to the modelling assumptions.
2. It may mask excessive random noise in cases where the random realization of the systematic error is close to zero, but the uncertainty budget is inflated. Complementing the test by the RIP and RIPQC methodology can reduce the risk of this.

Generally, this testing framework can be utilized as an Inclination Difference Test (IDT), Azimuth Difference Test (ADT) or Coordinate Difference Test (CODT). To fully meet the zero-mean assumption of the test, a sufficient randomization of the systematic contributions is required and could be achieved using results from different tools and runs in one single test. This in turn would render CODT impossible and cannot be applied to give insights into the performance of a single tool. Thus, a set of tests that can gauge the systematic and random uncertainty performances separately and perform coordinate difference χ^2 -tests in the presence of systematic uncertainty contributions is desirable.

Table 3—Overview of survey comparison results employing RIP (Ali 2020, Naschenveng 2023) and χ^2 -testing methodologies. Shown are the results of the standard RIP test, the RIPQC test, the standard χ^2 -test on differences in inclination and magnetic azimuth, the standard χ^2 -test on the coordinate differences in HL(A), and the non-central χ^2 -test on the lateral coordinate differences to highlight the successful separation of systematic and random contributions. All χ^2 -test results constitute a positive result if the numeric value does not exceed one. Note that the along-hole coordinate A is not shown as any differential depth error is transferred to the other coordinates by the interpolation procedure. A * marks the on-demand validation for well 3.

Well	Sect.	RIP Avg.		RIP Std.		RIPQC Syst.		RIPQC Rand.		χ^2		χ^2 HL(A)		NC- χ^2
		Inc	Azi	Inc	Azi	Inc	Azi	Inc	Azi	Inc	Azi	H	L	
2	1	Good	Good	Good	Good	Pass	Pass	Pass	Pass	0.03	0.21	0.02	0.37	0.32
2	2	Good	Good	Avg.	Good	Pass	Pass	Pass	Pass	0.04	0.28	0.02	0.42	0.37
2	3	Good	Good	Good	Good	Pass	Pass	Pass	Pass	0.12	0.12	0.11	0.07	0.13
2	4	Good	Good	Avg.	Good	Pass	Pass	Pass	Pass	0.06	0.31	0.01	0.40	0.48
2	5	Good	Good	Good	Good	Pass	Pass	Pass	Pass	0.07	0.22	0.03	0.18	0.28
2	6	Good	Good	Good	Good	Pass	Pass	Pass	Pass	0.18	0.17	0.01	0.05	0.11
3	1*	Good	Good	Good	Good	Pass	Pass	Pass	Pass	0.39	0.36	0.68	0.03	0.08
3	2	Good	Good	Avg.	Good	Pass	Pass	Pass	Pass	0.04	0.40	0.04	1.07	0.92
3	3	Good	Good	Poor	Good	Pass	Pass	Pass	Pass	0.08	0.43	0.03	0.92	0.71
3	4	Good	Good	Poor	Good	Pass	Pass	Pass	Pass	0.05	0.45	0.01	1.07	0.87
3	5	Good	Good	Poor	Good	Pass	Pass	Pass	Pass	0.07	0.36	0.02	0.52	0.49
3	6	Good	Good	Avg.	Good	Pass	Pass	Pass	Pass	0.05	0.27	0.03	0.19	0.28
3	7	Good	Good	Avg.	Good	Pass	Pass	Pass	Pass	0.39	0.41	0.05	0.66	0.69

Non-Central χ^2 -Tests (NC- χ^2)

The non-central χ^2 -test differs from its central counterpart by dropping the mean-free assumption of the underlying normal distribution. To test against it, an estimation of the mean of the underlying normal distribution is required, that can conveniently be provided by the mean calculations of the RIP testing. In cases where the central χ^2 -test fails due to the survey data not conforming to the mean-free assumptions that the test implies, the non-central χ^2 -test can be able to provide a more nuanced insight towards the conformity of the presented survey data with error modelling assumptions. It is crucial, that the non-central χ^2 -test only considers the modelling of random contributions to the observed difference, as the systematic contributions are already accounted for by utilizing the mean estimation.

Error Term Considerations when comparing SSAMWD and CRSAM surveys

Independent of the testing method, attention must be paid to possible correlations between the error sources present in the survey types under comparison. Any error source contributing to both survey types in a systematic manner, cannot be raised to explain observed differences. Especially when comparing surveys acquired by the same sensors in the same tool under different operating and algorithmic conditions, this elimination of correlated error sources is a crucial task to not provide an inflated uncertainty budget, resulting in overly optimistic test results, i.e. false positives. [Table 4](#) in [Appendix A](#) shows the currently assumed error sources of the static ISCWSA MWD Rev4 error model ([ISCWSA 2015](#)) and the release candidate of the ISCWSA Continuous Rotating 6-Axis Mag. error model ([Willerth and Akklestad 2024](#), [ISCWSA 2025c](#)). The two survey mode associated uncertainty models collectively account for 37 error sources, 19 of them mutual. Among these are most noticeably all error sources rooted in the limited knowledge of the exact reference field values for magnetic dip and declination (DECG, DECR, DBHG, DBHR) that are correlated for surveys acquired on the same run. All axial interference (AMIL) and axial sensor contributions (ABZ, ASZ, MBZ, MSZ) are also considered fully correlated, as this measurement axis is left unaffected by changing the survey mode. All 18 error sources that are mutually exclusive to either survey type will fully contribute to any difference observed when comparing the survey results. Only a single error source is considered mutual and random and thus contributing through both surveys. Influences from gravitational sag (SAG) are considered correlated between the survey modes. The treatment of the misalignment contributions (XYM1, XYM2, XYM3, XYM4) is broken down into a static contribution and rotating contribution ([Williamson 2000](#)) (see [Appendix B](#)). While the static portion can be attributed to both surveys, the rotational portion is averaged and suppressed in the CRSAM operating mode, leaving only the rotating misalignment in the SSAMWD as a mutually exclusive contribution that is assumed randomized in the survey comparison due tool face variation between survey stations. Similarly, due to their strong suppression in the CRSAM survey calculation, the cross-axial sensor biases only enter through the SSAMWD survey into the comparison, under a tool face randomization assumption. The cross-axial sensor scale errors are partially matched between the survey types, as the first virtual error sources, i.e. ASXY-TI1 and MSXY-TI1, are present in both survey types, albeit they carry the codes ASXY-ROT and MSXY-ROT, respectively, and are linked through their mutual weighting functions. All error sources unique to the CRSAM operation mode, such as drilling noise terms (AN1, AN2, ANZ), residual centrifugal acceleration terms (CA1, CA2), filter attenuation terms (AXY-ATTEN, MXY-ATTEN), phase shift (PSD) and eddy current terms (EDDY) and depth shift terms (DSC) fully contribute to the calculation of the expected survey deviations.

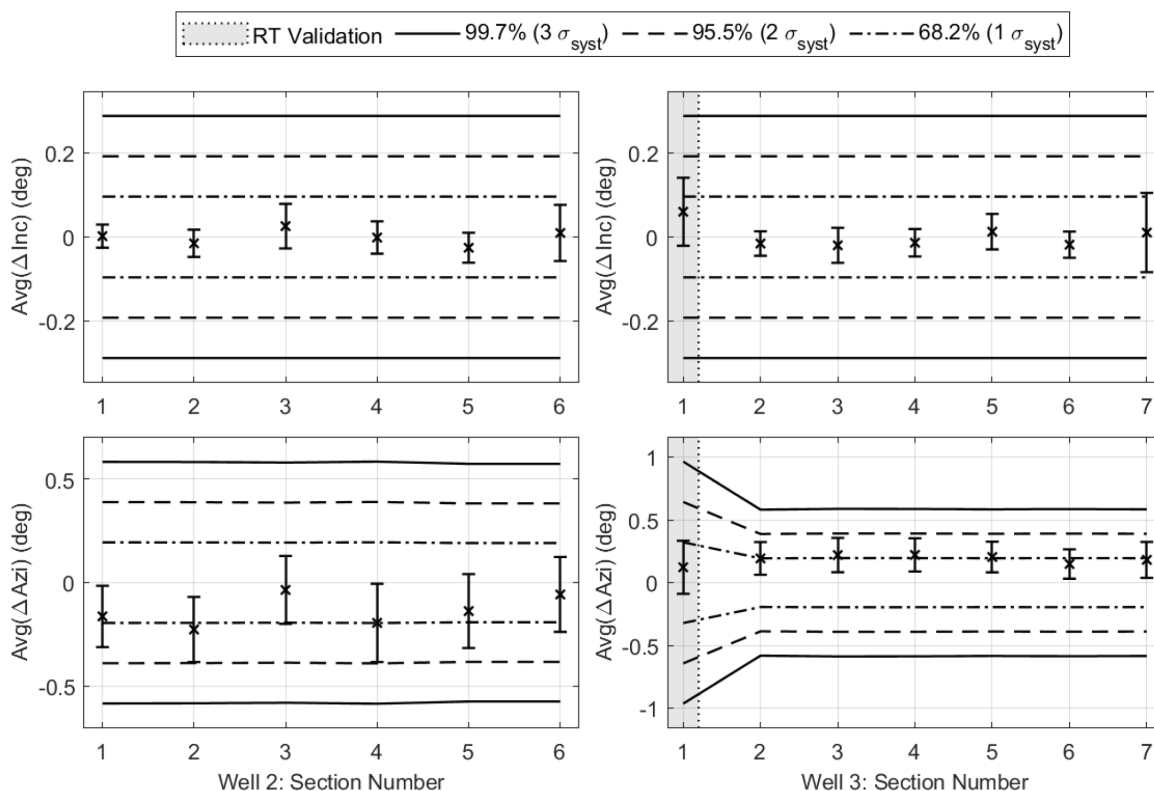


Figure 2—RIP testing of systematic deviations in the inclination differences ΔInc and azimuth differences ΔAzi for all sections on well number 2 and on well number 3. The results agree within the respective margins with expectations set by the uncertainty modelling assumptions for systematic error sources. In all cases the 99.7% confidence range of the estimations of the systematic deviations are fully included in the 95.5% margins of the modelling expectations, providing a quantitative agreement. Section 1 of Well 3 was used for on-demand validation (grey shading).

Survey Comparison Setup

The survey comparisons divide the survey log into distinct sections of 15 static survey stations along the horizontal of about 450 meters. In both cases (well 2 and 3) about 90 static surveys are available in the horizontal section until reaching TD, yielding six distinct sections per well for analysis. For all sections the tool face distribution was evaluated to ensure sufficient tool face spread to support the randomization assumption made in the error term comparison. The section 1 of well 2 was used for an on-demand validation, covering the full build of the wellbore in about 20 stations.

Well 2: Post-run validation of CRSAM performance

The second well was drilled by relying on SSAMWD survey as a definitive survey posted by the directional driller. CRSAM were acquired and collected at surface for a post-validation against the static surveys. All six sections pass all applied methodologies without any noticeable need for refinement. The results are summarized in Table 3 and visualized in Figures 2 and 3, as well as in Figures 4 and 5 in Appendix C. This successful performance verification showcased the comparison methodology, paving the way for the successive deployment of the on-demand validation and use of CRSAM surveys as definitive surveys on Well 3.

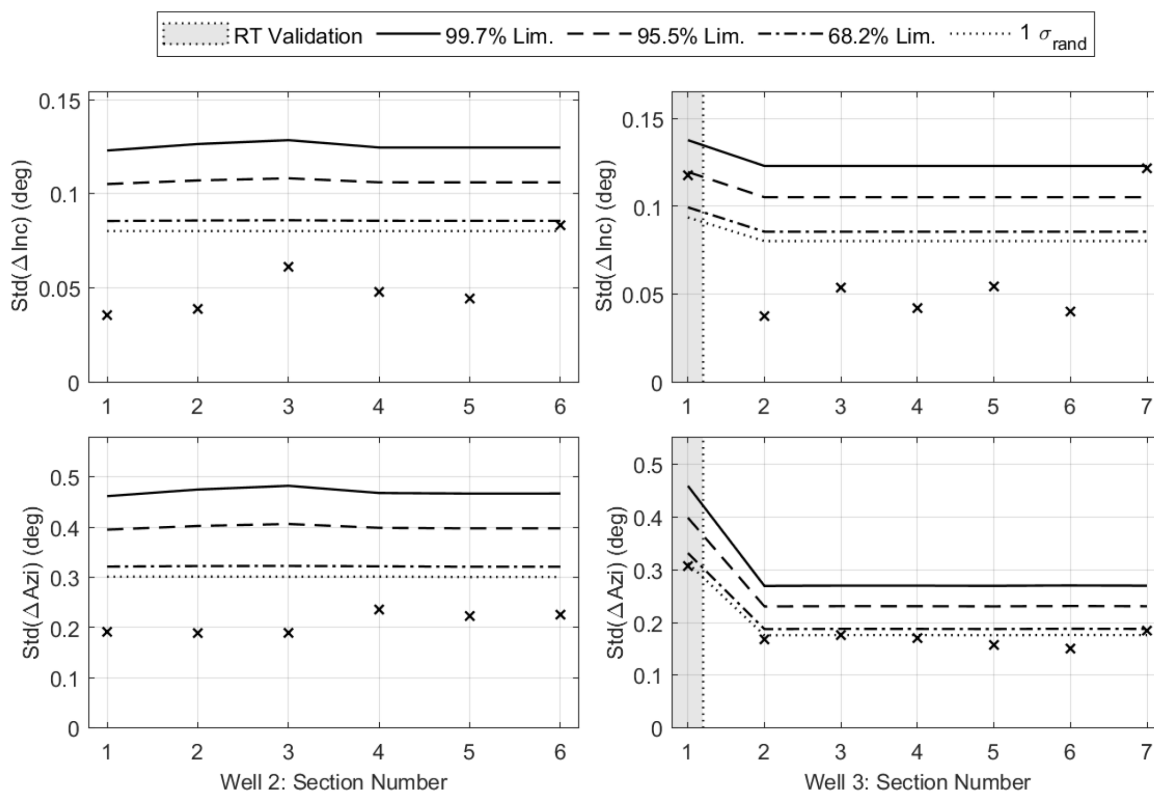


Figure 3—RIP testing of random deviations in the inclination differences ΔInc and azimuth differences ΔAzi for all sections on well number 2 and on well number 3. The results agree within the respective margins with expectations set by the uncertainty modelling assumptions for random error sources. In all cases the 99.7% confidence range of the estimations of the systematic deviations are fully included not only in the 99.7% margins of the modelling expectations, providing a quantitative agreement. Section 1 of Well 3 was used for on-demand validation (grey shading). Note that the expression of random noise sources can vary along the wellbore and might be section dependent. The error modelling assumptions must correctly capture scenarios of elevated drilling noise under which a CRSAM survey might be acquired.

Well 3: On-demand validation of CRSAM and post-run validation in sections

The third well was drilled relying on SSAMWD surveys as a definitive survey for the vertical section and throughout the build section up until three surveys stations into the horizontal section, while CRSAM surveys were acquired and collected at surface. At this point in time the survey log was sent in for analysis, with respect to the differential performance of the SSAMWD and CRSAM survey modes. Within the hour the result of the analysis was returned to the rig crew with all tests passing in section 1 of well 3 (see Table 3, Figures 2 and 3, as well as Figures 6 and 7 in Appendix C), signaling a positive go-ahead. Consecutively, for the remainder of the run up until reaching TD, the CRSAM surveys were used as the definitive survey. Automatically acquired SSAMWD surveys were still received on surface and stored. The post-run validation of the remaining six sections of the well shows good agreement of the survey data with the error modelling assumptions but highlights the inability of the coordinate difference central χ^2 -test to reliably assess the differential tool performance with respect to the appropriate error models in the presence of systematic errors, even though these are considered and accounted for when applying PUM calculations. The non-central χ^2 -test enables a more nuanced analysis using prior knowledge about systematic deviations and jointly with the RIPQC test enables a holistic judgment of the comparative tool performance with respect to random and systematic error sources separately (see Table 3, Figures 2 and 3, as well as Figures 8 and 9 in Appendix C).

The utilization of CRSAM as a definitive survey leads to direct time savings removing any additional waiting time for survey confirmation on connection, in this specific approach on 9 out of ten stands. For MWD tools that operate without a battery time savings would increase as the additional waiting time for the tool power on could be avoided as well. However, the benefit of a battery powered SSAMWD survey

acquisition becomes evident when reviewing the presented approach through the lens of managing change, as it will always provide retrievable and reliable directional survey data that can be compared against, easing the introduction of CRSAM in this specific instance. Additional invisible time savings might be found from an improvement in whole quality thanks to more frequent reliable survey data enabling a smoother wellbore, resulting in faster casing runs.

Observations and Conclusions

The successful completion of the test program can be summarized in the completion of key milestones towards full capitalization on the operational benefits offered by CRSAM surveying (Table 1). First and foremost, it marks the introduction of CRSAM technology in joint operations, with the specific highlight of an on-demand PUM and survey cross-validation within the hour, enabled by proper planning and involvement between local teams and headquarters. The validation procedure encompassed the utilization of an extended RIP test and a χ^2 -test to confirm congruency of both survey types for the specific tool run. This confirmation through a comparison to SSAMWD surveys enabled the remainder of the run to rely on CRSAM surveys as the definitive survey as a first in joint operations and to wait for the acquisition of precautionary static surveys on every 10th connection, to continuously verify the agreement between CRSAM surveys and SSAMWD surveys. Beyond this, the automated directional drilling was able to successfully deliver directional drilling advice that was acted upon by the rig crew based on posted CRSAM surveys for the first time, paving the way for on-arrival use of high frequency CRSAM surveys for automated directional drilling services.

More generally, the test did demonstrate the typical CRSAM survey performance, helped gain trust, and improved insight into the associated uncertainty quantification. The presented approach demonstrates that carefully managing change aids the introduction of new technologies, that help save time, reduce cost, and increase wellbore resolution. while maintaining the operational integrity of uncertainty calculations, required to successfully place the well, by meeting geological targets and excluding HSE risks.

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References

- Ali, T., Ledroz, A., Weston, J. and Allen, W., 2020. Validation of Error Models – A Key Component of Risk Mitigation in Wellbore Collision Challenges, IADC/SPE paper 199554-MS, IADC/SPE International Drilling Conference and Exhibition, Galveston, Texas, 3-5 March 2020.
- Dreisig, H. and Reynaud, D., 2023. Comparison of Stationary Versus Non-Stationary MWD Surveys to Justify a Pragmatic Use of Non-Stationary MWD Surveys, Presentation given at the 58th ISCWSA meeting, San Antonio, Texas, 19 October 2023.
- Ekseth, R., Torkildsen, R., Brooks, A., Weston, J., Nyrnes, E., Wilson, H., and Kovalenko, K., 2007. High-Integrity Wellbore Surveys: Methods for Eliminating Gross Errors, SPE/IADC 105558, SPE/IADC Drilling Conference, Amsterdam, The Netherlands, 20–22 February 2007
- Elgizawy, M., Lowdon, R., Edmunds, M., Breen, M. and Ekwue, W., 2020. Accuracy Prediction of Zero-Survey Time Definitive Dynamic MWD Surveys, SPE paper 203361-MS, Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAE, 9 November 2020. Also presented at 52nd ISCWSA meeting, Remote Online, 22 October 2020.
- Hansen, C., Stokes, M., Miting, R., Quattrone, F., Klemme, V., Nageshwara Rao, K., Wassermann, I., and Zaeper, R., 2020. Automated Trajectory Drilling for Rotary Steerable Systems, IADC/SPE paper 199647-MS, IADC/SPE International Drilling Conference and Exhibition, Galveston, Texas, 3-5 March 2020.

- Herrera Anda, D., Hansen, C., Hueper, A., Koenecke, A., Gjertsen, M., Pubanz, S., and Vogt, A., 2024. First Successful Field Test of Continuous Surveying While Drilling in Slim Hole, SPE paper 222343-MS, ADIPEC, Abu Dhabi, UAE, November 4–7, 2024
- ISCWSA, 2015. ISCWSA MWD Error Model Revision 4, <https://www.iscwsa.net/articles/iscwsa-mwd-error-model-revision-4/> (accessed 25 August 2025)
- ISCWSA, 2019. *Error Model Documentation, Latest Version - Revision 5*, <https://www.iscwsa.net/error-model-documentation/> (accessed 25 August 2025)
- ISCWSA, 2025. *About The Industry Steering Committee on Wellbore Survey Accuracy*, <https://www.iscwsa.net/about/>, paragraph 3 (accessed 25 August 2025)
- ISCWSA, 2025. *Error Model Sub-Committee*, <https://www.iscwsa.net/committees/error-model/>, paragraph 5 (accessed 25 August 2025)
- ISCWSA, 2025. Release Candidate Rotating 6-axis MWD+SRGM Error Model, Error Model Sub-Committee, Rotating Error Model Work Group, <https://www.iscwsa.net/files/998/> (accessed 25 August 2025)
- Lowdon, R., Breen, M., Mouyiasis, M., Edmunds, M., Bulychenkow, K. and Brovko, K., 2019. Eliminating Rig Time from MWD Surveying", IADC/SPE paper 194057-MS IADC/SPE International Drilling Conference and Exhibition, The Hague, The Netherlands, 5-7 March 2019.
- Naschenveng, A., Isaac, L., Laroca, F., Franceschi, C., Neves, A., Canté, F., Ledroz, A., and Carrasquilla, A., 2023. Definitive Survey Methodology – Update to an Old Concept for Higher Reliability, IADC/SPE-212492-MS, IADC/SPE International Drilling Conference and Exhibition, Stavanger, Norway, 7 – 9 March 2023
- Torres Cabrera, N., Kolton, L., Pozo, M., Grymalyuk, S., Macedo, R. R., Ait Ali, R., and Forshaw, M., 2024. Hybrid Edge-Cloud Automated Advisory System for Well Construction Ensures Well Delivery While Improving Drilling Performances, SPE paper 222445-MS, ADIPEC, Abu Dhabi, UAE, November 4–7, 2024
- Willerth, M., 2024. Error Model SubCommittee Update, Non Static Surveys check-in, <https://www.iscwsa.net/media/files/events-event/538d25cd/i59-2-231214-errormodelmeeting.pdf>, page 4, Presentation given at the 59th ISCWSA meeting, Glasgow, Scotland, 17-18 April 2024
- Willerth, M., and Akklestad, D., 2024. Error Model Maintenance and Collision Avoidance Joint Sub-committee Update, slides 4 and 5, <https://www.iscwsa.net/media/files/events-event/f5ea9afa/9-iscwsa-60th-error-model-and-ca-update.pdf>, Presentation given at the 60th ISCWSA meeting, New Orleans, Louisiana, 25-26 September 2024
- Williamson, H. S., 2000. Accuracy prediction for directional MWD, SPE paper 67616, *SPE Drilling & Completion* **15** (4)

Appendix A

Error Term Comparison Table

Table 4—Error term comparison between SSAMWD and CRSAM surveys acquired by the same directional MWD tool. The contribution columns list a scaling factor that is applied to modify the original term magnitude to capture partial correlations.

Code	Alt. Code	Term Description	Wt.Fn.	Type	SSAMWD	CRSAM	Shared	Corr.	SSAMWD Contr.	CRSAM Contr.	Mode
SAG		MWD: Sag	SAG	Align	yes	yes	yes	Full	0	0	-
XYM1		XY Misalignment 1	XYM1	Align	yes	yes	yes	Partial	0.75	0	R
XYM2		XY Misalignment 2	XYM2	Align	yes	yes	yes	Partial	0.75	0	R
XYM3		XY Misalignment 3	XYM3	Align	yes	yes	yes	Partial	0.75	0	R
XYM4		XY Misalignment 4	XYM4	Align	yes	yes	yes	Partial	0.75	0	R
DECG		MWD: Declination - Global	AZ	AziRef	yes	yes	yes	Full	0	0	-
DECR		MWD: Declination - Random	AZ	AziRef	yes	yes	yes	Full	0	0	-
DBHG		MWD: BH-Dep. Decl. - Global	DBH	AziRef	yes	yes	yes	Full	0	0	-
DBHR		MWD: BH-Dep. Decl. - Random	DBH	AziRef	yes	yes	yes	Full	0	0	-
DRFR		Depth: Depth Ref. - Random	DREF	Depth	yes	yes	yes	Full	1	1	R
DSFS		Depth: Depth Scale - Systematic	DSF	Depth	yes	yes	yes	Full	0	0	-
DSTG		Depth: Depth Stretch - Global	DST	Depth	yes	yes	yes	Full	0	0	-
AMIL		MWD: Axial Interf.	AMIL	Mgntcs	yes	yes	yes	Full	0	0	-
EDDY		MWD: XY-Interf. Eddy Currents	EDDY	Mgntcs	no	yes	no	None	0	1	S
ABXY-TI1R		MWD TF Ind: X & Y Acc. Bias	ABXY-TI1	Sensor	yes	no	no	None	1	0	R
ABXY-TI2R		MWD TF Ind: X & Y Acc. Bias	ABXY-TI2	Sensor	yes	no	no	None	1	0	R
ABZ		MWD: Z-Acc. Bias	ABZ	Sensor	yes	yes	yes	Full	0	0	-
ASXY-TI1	ASXY-ROT	MWD TF Ind: X & Y Acc. Scale	ASXY-TI1	Sensor	yes	yes	yes	Full	0	0	-
ASXY-TI2R		MWD TF Ind: X & Y Acc. Scale	ASXY-TI2	Sensor	yes	no	no	None	1	0	R
ASXY-TI3R		MWD TF Ind: X & Y Acc. Scale	ASXY-TI3	Sensor	yes	no	no	None	1	0	R
ASZ		MWD: Z-Acc. Scale	ASZ	Sensor	yes	yes	yes	Full	0	0	-
MBXY-TI1R		MWD TF Ind: X & Y Mag. Bias	MBXY-TI1	Sensor	yes	no	no	None	1	0	R
MBXY-TI2R		MWD TF Ind: X & Y Mag. Bias	MBXY-TI2	Sensor	yes	no	no	None	1	0	R
MBZ		MWD: Z-Magnetometer Bias	MBZ	Sensor	yes	yes	yes	Full	0	0	-
MSXY-TI1	MSXY-ROT	MWD TF Ind: X & Y Mag. Scale	MSXY-TI1	Sensor	yes	yes	yes	Full	0	0	-
MSXY-TI2R		MWD TF Ind: X & Y Mag. Scale	MSXY-TI2	Sensor	yes	no	no	None	1	0	R
MSXY-TI3R		MWD TF Ind: X & Y Mag. Scale	MSXY-TI3	Sensor	yes	no	no	None	1	0	R
MSZ		MWD: Z-Magnetometer Scale Factor	MSZ	Sensor	yes	yes	yes	Full	0	0	-
AN1		MWD: XY-Shock and Vibe, Term 1	AN1	Sensor	no	yes	no	Full	0	1	R
AN2		MWD: XY-Shock and Vibe, Term 2	AN2	Sensor	no	yes	no	Full	0	1	R
ANZ		MWD: Z-Shock and Vibe	ANZ	Sensor	no	yes	no	Full	0	1	R
AXY-ATTEN		MWD: Acc. XY-Att. LP Filter	AXY-ATTEN	Sensor	no	yes	no	Full	0	1	S
MXY-ATTEN		MWD: Mag. XY-Att. LP Filter	MXY-ATTEN	Sensor	no	yes	no	Full	0	1	S
AMXY-PS		MWD: XY-Sensor Phase Shift	AMXY-PS	Sensor	no	yes	no	Full	0	1	S
CA1		MWD: XY-Centr. Accel, Term 1	CA1	Sensor	no	yes	no	Full	0	1	R
CA2		MWD: XY-Centr. Accel, Term 2	CA2	Sensor	no	yes	no	Full	0	1	R
DSC		MWD: Depth Shift Comp.	DSC	Sensor	no	yes	no	Full	0	1	S

Appendix B

Residual Misalignment Contribution Heuristic

The misalignment terms in the ISCWSA error models historically are utilized to cover and lump sum contributions from multiple layered error sources that all contribute to an effective misalignment of the MWD sensing directions with respect to the central borehole axis. Among the error sources are residual misalignment of physical sensor axis, of directional probes within their respective collar and of the collar with respect to the borehole direction ([Williamson 2000](#)). In the original formulation a misalignment uncertainty 1-sigma magnitude of 0.06° was assumed, of which a statistically independent contribution of 0.046° was considered to be rotating with the tool, while 0.04° were considered fixed with respect to the borehole. The magnitudes of the misalignment terms have been scrutinized over the iterations of the ISCWSA MWD generic error model publications leading to values of 0.1° in the ISCWSA MWD Rev. 4 (ISCWSA 2015) and even a partial increase to 0.3° for the virtual terms covering vertical sections (XYM3, XYM4) in ISCWSA MWD Rev. 5 (ISCWSA 2019). While this adoption presents an increase in uncertainty ranges for vertical sections, the presented analysis is predominantly performed on data acquired in horizontal sections and would only be marginally affected by adopting this change, that would constitute a relaxation of the testing conditions. Heuristically extending the underlying separation from ([Williamson 2000](#)) to cover the magnitude of 0.1° leads to a static misalignment contribution of $(0.1^\circ)(0.06^\circ) \times 0.04^\circ \approx 0.067^\circ$ leaving a portion of $\sqrt{(0.1^\circ)^2 - (0.067^\circ)^2} \approx 0.075^\circ$ to rotate with the tool. It is the rotating contribution that is considered in the error model comparison (see [Appendix A, Table 3](#)) and considered contributing to survey differences.

Appendix C

Additional Figures

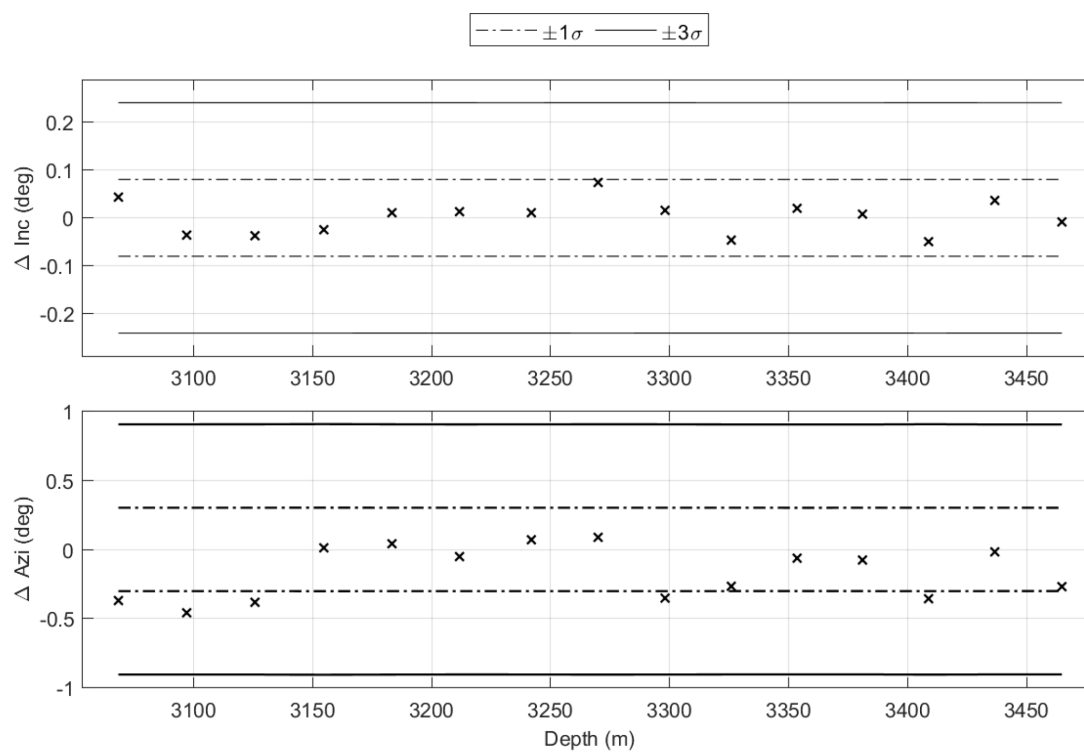


Figure 4—Example of observed inclination differences ΔInc and azimuth differences ΔAzi for section 1 on well number 1 during the post-run validation step. The results agree within the respective collective margins with expectations set by the uncertainty modelling assumptions. These values are used as in put for RIP and χ^2 -testing. Spacing of datapoints is set by the availability of SSAMWD surveys.

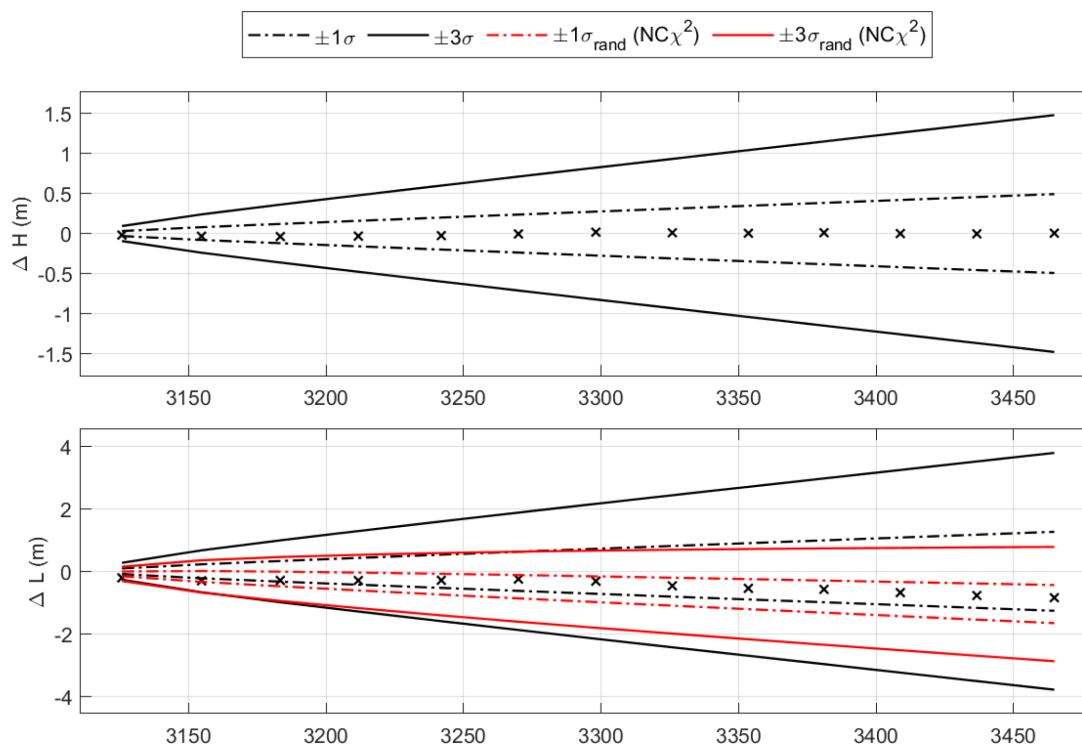


Figure 5—Example of observed positional differences in high-side coordinates ΔH and lateral coordinates ΔL for section 1 on well number 1 during the post-run validation step. The results agree within the respective collective margins with expectations set by the uncertainty modelling assumptions. These values are used as in put for χ^2 -testing. Spacing of datapoints is set by the availability of SSAMWD surveys. Note that the along-hole coordinate difference ΔA is not shown as any differential depth error is transferred to the other coordinates by the interpolation procedure.

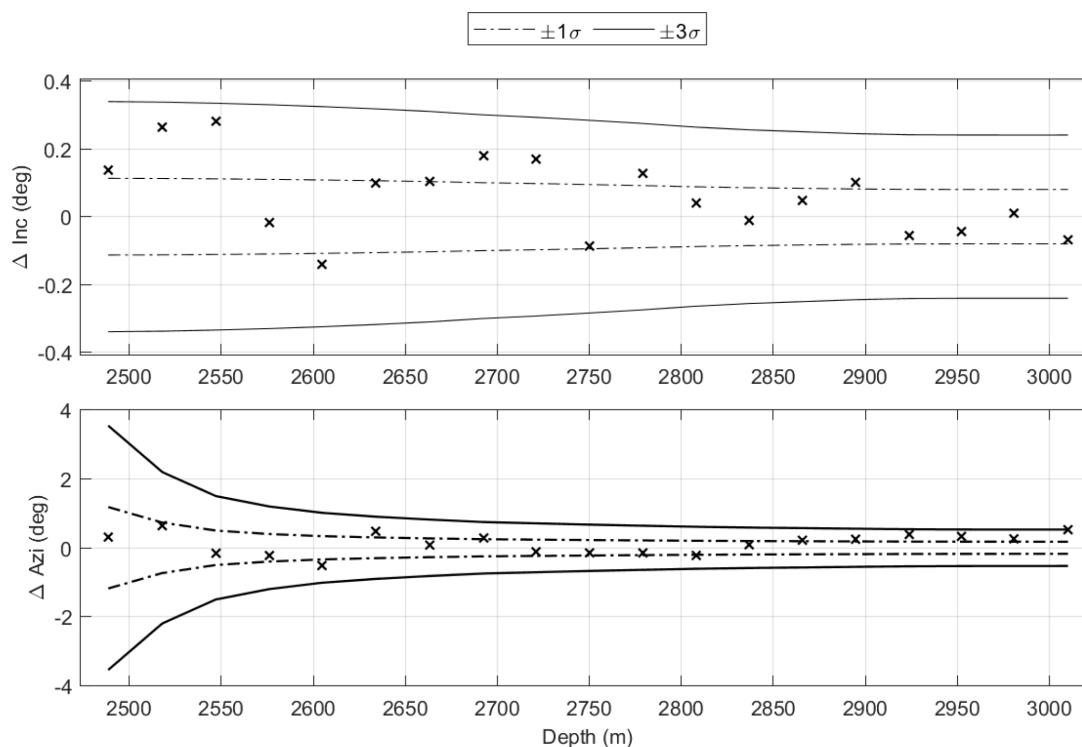


Figure 6—Observed inclination differences ΔInc and azimuth differences ΔAzi for section 1 on well number 2 during the on-demand validation step. The results agree within the respective collective margins with expectations set by the uncertainty modelling assumptions. These values are used as in put for RIP and χ^2 testing. Spacing of datapoints is set by the availability of SSAMWD surveys.

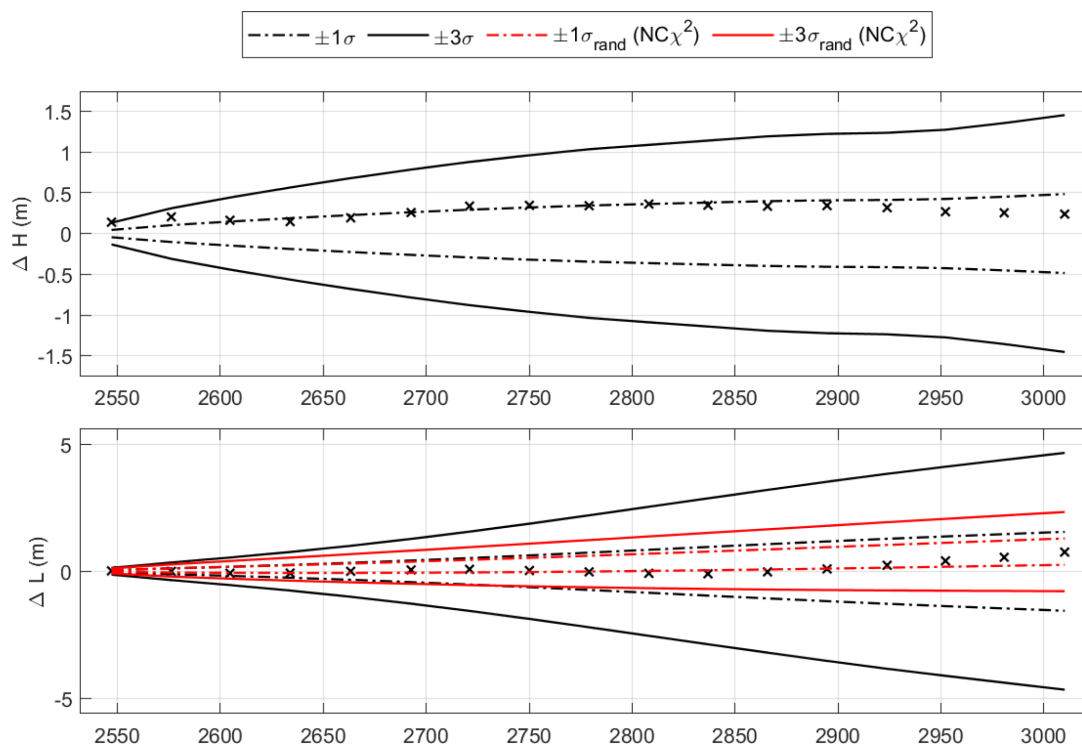


Figure 7—Observed positional differences in high-side coordinates ΔH and lateral coordinates ΔL for section 1 on well number 2 during the on-demand validation step. The results agree within the respective collective margins with expectations set by the uncertainty modelling assumptions. These values are used as in put for χ^2 -testing. Spacing of datapoints is set by the availability of SSAMWD surveys. Note that the along-hole coordinate difference ΔA is not shown as any differential depth error is transferred to the other coordinates by the interpolation procedure.

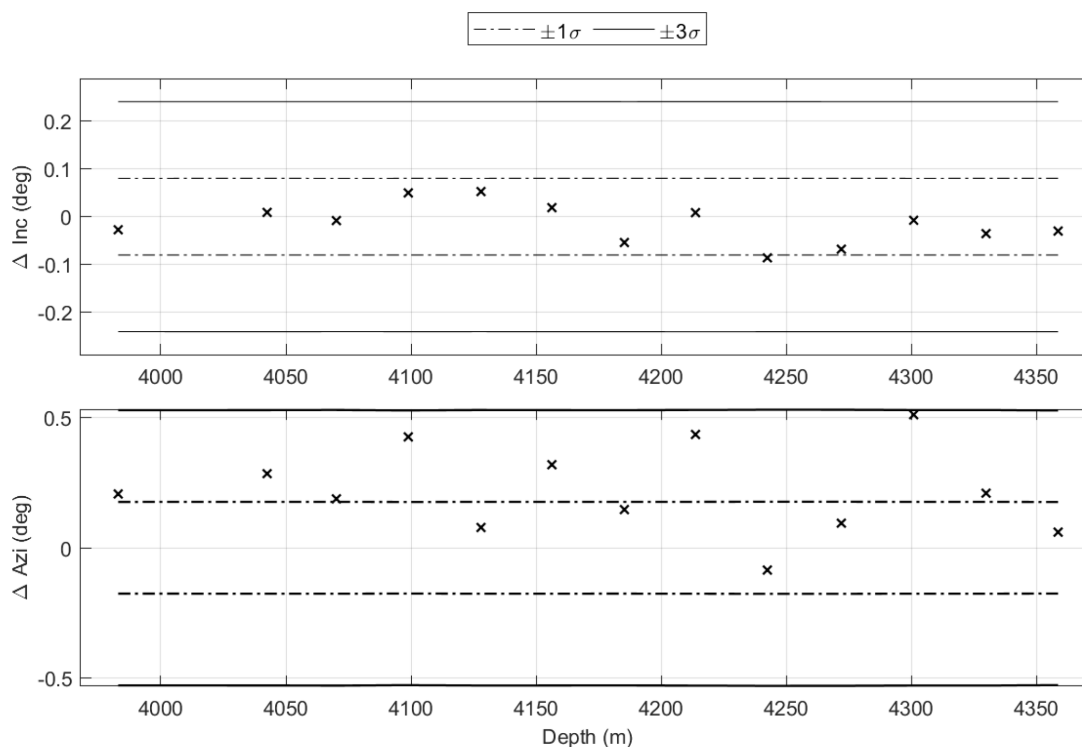


Figure 8—Observed inclination differences ΔInc and azimuth differences ΔAzi for section 4 on well number 2 during the post-run validation step. The results agree within the respective collective margins with expectations set by the uncertainty modelling assumptions but express a systematic azimuth deviation that will to a failing coordinate difference χ^2 -test in the lateral coordinate differences ΔL . Spacing of datapoints is set by the availability of SSAMWD surveys.

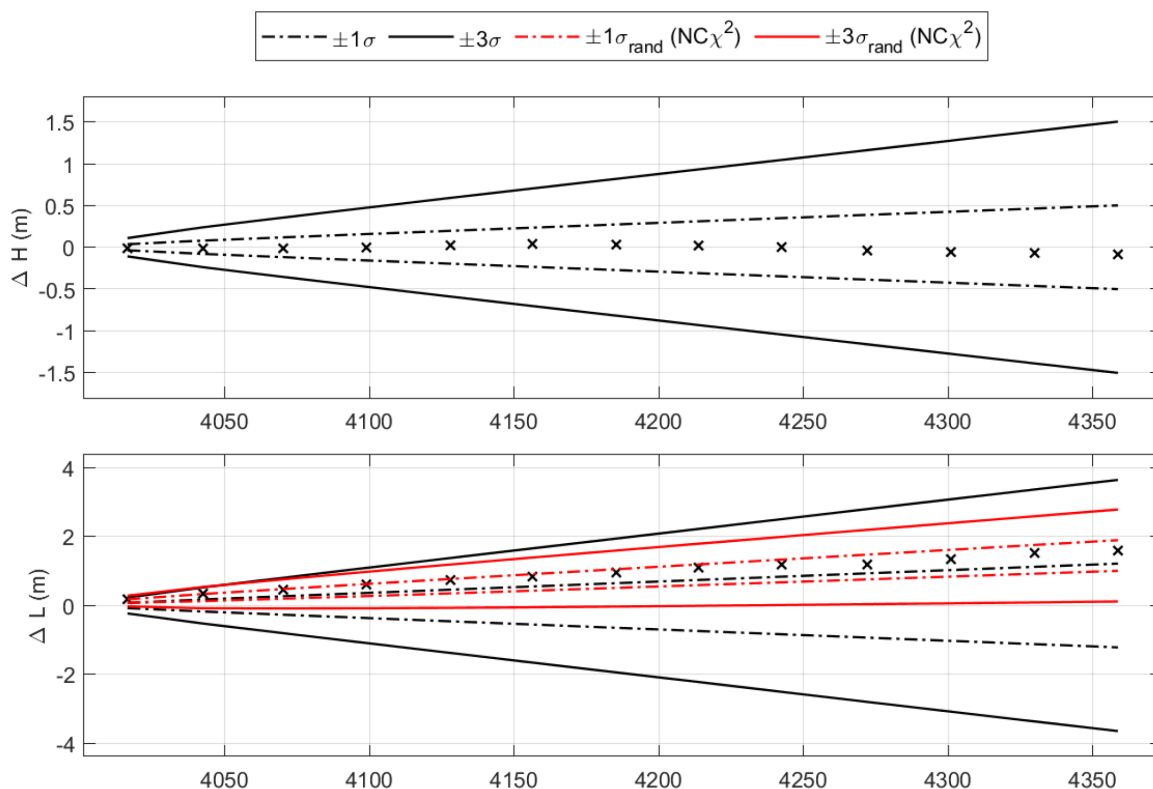


Figure 9—Observed positional differences in high-side coordinates ΔH and lateral coordinates ΔL for section 4 on well number 2 during the post-run validation step. The results fail to agree within the margins set by the central χ^2 test for the lateral coordinate difference ΔL due to the lacking capability of the test to consider systematic uncertainty assumptions. By employing a non-central χ^2 -test the systematic effects are accurately captured and in agreement with the error modelling assumptions. Spacing of datapoints is set by the availability of SSAMWD surveys. Note that the along-hole coordinate difference ΔA is not shown as any differential depth error is transferred to the other coordinates by the interpolation procedure.